

Assessment of Alzheimer's Disease Progression Using a Comorbidity-Aware Meta-Bayesian Framework with Resting-State fMRI

Second Review

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Outline

- ① First review – queries and answers
- ② Problem statement and objectives
- ③ Dataset
- ④ Proposed system architecture
- ⑤ Modules and methodology
- ⑥ Core processing logic
- ⑦ Implementation status
- ⑧ Results and analysis
- ⑨ Plan for the remaining work
- ⑩ References

First Review – Queries and Answers

Query	Answer
[Query 1 raised by the panel]	[Answer]
[Query 2 raised by the panel]	[Answer]
[Query 3 raised by the panel]	[Answer]

Design Revisions Since Review I

- ▶ **Feature matrix is 246×3 , not 246×4 :** FC strength = DC/245 exactly ($r = 1.0$), so it adds no information.
- ▶ **Validation unit is the subject:** 165 scans come from only **127 subjects**; MCI has 32 scans but just 14 subjects.
- ▶ **No AD scan is unrecoverable:** the 6 scans once listed as lost were found and processed (25/25).
- ▶ **Stage order fixed:** CN $\rightarrow$ SMC $\rightarrow$ EMCI $\rightarrow$ MCI $\rightarrow$ LMCI $\rightarrow$ AD.
- ▶ **Comorbidities named honestly:** ADNI history gives broad cardiovascular / endocrine categories, not “hypertension” or “diabetes”.

Problem Statement

Problem

Most rs-fMRI studies only separate **AD from healthy controls**. Clinicians need to know **where on the six-stage continuum** a person is, with an honest **measure of confidence**.

Task

- ▶ Input: rs-fMRI scan(s) + comorbidity record
- ▶ Output: ordinal stage $y \in \{\text{CN, SMC, EMCI, MCI, LMCI, AD}\}$, its probability distribution and uncertainty

Constraints

- ▶ BOLD only: no T1w image, no slice timing
- ▶ Mixed TRs (3.0 s, 2.25 s, 6.02 s)
- ▶ 127 subjects, as few as 14 per class

Project Objectives

#	Objective	Status
1	To curate an analysis-ready ADNI rs-fMRI cohort over six stages, documenting every exclusion	Completed
2	To build a reproducible, BOLD-only preprocessing pipeline with per-scan QC	Completed
3	To extract and independently verify ALFF, ReHo, FC and DC on the Brainnetome-246 atlas	Completed
4	To produce a leakage-free , model-ready feature set	Completed
5	To develop a comorbidity-aware Meta-Bayesian ordinal model with uncertainty	Planned
6	To evaluate it under subject-grouped cross-validation and explain its predictions	Planned

Dataset: ADNI Resting-State fMRI

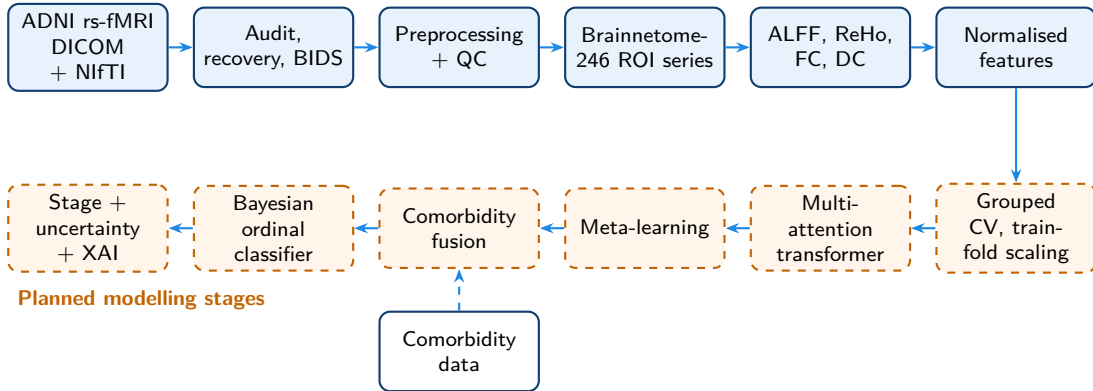
Group	Eligible	Processed	Subjects
CN	29	27	26
SMC	32	31	21
EMCI	30	29	26
MCI	35	32	14
LMCI	22	21	20
AD	25	25	20
Total	173	165	127

Accounting

- ▶ 175 raw scans audited
- ▶ 29 recovered from raw DICOM
- ▶ 2 excluded (truncated, duplicate)
- ▶ 8 rejected: $TR = 6.02\text{ s}$ breaks the 0.10 Hz band (Nyquist 0.083 Hz)

Proposed System Architecture

Implemented and verified (165/165 scans)



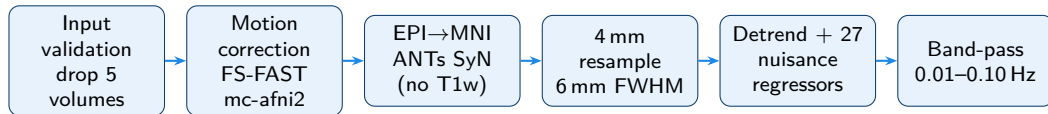
- 1 **Module 1 – Data audit and curation:** inventory, DICOM recovery (dcm2niix), duplicate audit, exclusions, slice-timing investigation.
- 2 **Module 2 – Preprocessing:** motion correction, EPI→MNI, smoothing, nuisance regression, band-pass, QC.
- 3 **Module 3 – Parcellation and biomarkers:** Brainnetome-246, ROI time series, ALFF, ReHo, FC, DC + independent checks.
- 4 **Module 4 – Normalisation:** within-subject mALFF, mReHo, DC_z , Fisher-z FC $\rightarrow 246 \times 3 + 246 \times 246$.
- 5 **Module 5 – Staging model (planned):** multi-attention transformer, meta-learning, comorbidity fusion, Bayesian ordinal head.
- 6 **Module 6 – Testing and evaluation:** QC, pilot reproduction gate, dataset audits; subject-grouped CV for the model.

- 1 Audit ADNI data, recover missing scans, document exclusions
- 2 Preprocess all eligible scans with one frozen pipeline
- 3 Align Brainnetome-246; extract ALFF, ReHo, FC, DC
- 4 Verify independently; normalise within subject
- 5 Build subject-grouped CV folds; scale on training folds only
- 6 Train transformer + meta-learning + fusion + ordinal head
- 7 Evaluate, calibrate and explain

Evaluation Metrics

- ▶ Balanced accuracy
- ▶ Mean absolute stage error
- ▶ Quadratic weighted κ
- ▶ Expected calibration error
- ▶ Ablations (no ordinal / no fusion / no meta-learning)

Core Processing Logic: Preprocessing Pipeline



Nuisance model (OLS)

Friston-24 motion + WM prior + global signal + intercept.

CSF prior failed the reliability gate in all 165 scans and was dropped, not replaced.

Never guessed

Slice timing: no metadata → not performed.
Nyquist check: $0.10 < 1/(2 TR)$, else the scan is rejected.

Core Processing Logic: Biomarkers and Normalisation

Biomarkers (per Brainnetome region)

- ▶ **ALFF**: mean FFT amplitude in 0.01–0.10 Hz
- ▶ **ReHo**: Kendall's W over a 27-voxel cube
- ▶ **FC**: 246×246 Pearson, signed, unthresholded
- ▶ **DC**: $DC_i = \sum_{j \neq i} FC_{ij}$

Within-subject normalisation

- ▶ $mALFF = ALFF_i / \overline{ALFF}$
- ▶ $mReHo = ReHo_i / \overline{ReHo}$
- ▶ DC_z : z-score within the scan
- ▶ $FC \rightarrow \text{artanh}(r)$, diagonal = 0

Model input per scan

246×3 regional matrix (mALFF, mReHo, DC_z) + 246×246 Fisher-z connectivity matrix

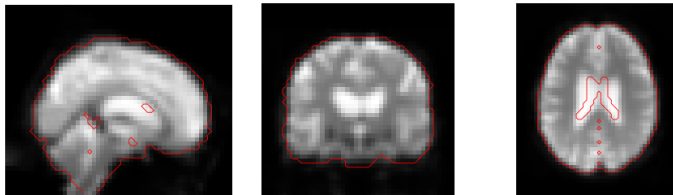
Implementation Status

Modules	Status	Evidence / Remarks
M1 Data audit and curation	Completed	175 audited, 29 recovered, 2 excluded
M2 Preprocessing + QC	Completed	165/173 scans; 8 rejected (TR 6.02 s)
M3 Parcellation + biomarkers	Completed	246/246 ROIs, 165/165 pass checks
M4 Normalisation	Completed	165/165, 0 failures
M5 Staging model	Planned	LMCI comorbidity tables pending
M6 Model evaluation	Planned	Subject-grouped CV protocol fixed

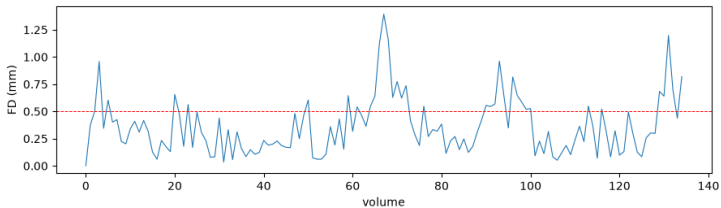
Stack: Python 3.12, NumPy, SciPy, NiBabel, Nilearn, ANTsPy, TemplateFlow, FreeSurfer 7.4.1 (WSL2); single worker with validated checkpoint/resume.

Implementation Output: Per-Scan QC (pilot AD subject)

sub-019S4549 ses-01 MNI registration

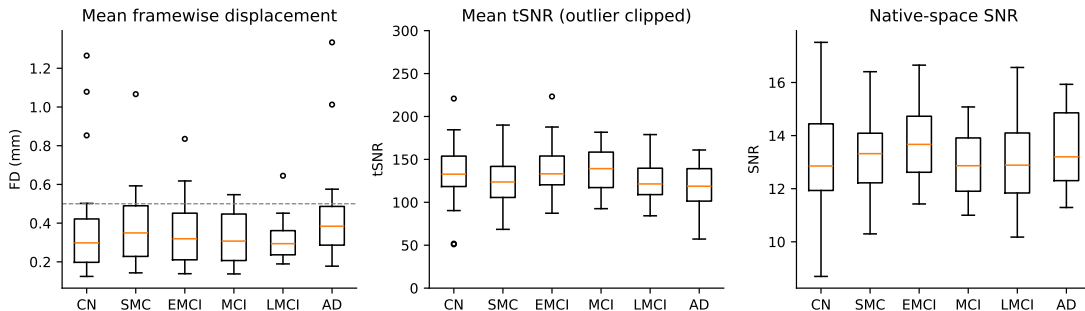


sub-019S4549 ses-01 FD



Top: MNI registration overlay. Bottom: framewise displacement (dashed line 0.5 mm).

Results: Preprocessing Quality Control



Key Findings

- ▶ Median tSNR 119–139 and SNR 12.9–13.8 across groups: no stage-ordered quality bias
- ▶ AD moves most: mean FD **0.44 mm**, 27.7% of volumes > 0.5 mm $\rightarrow$ motion used as a covariate

Results: Biomarker Verification (all 165 scans)

Check	Result
Scans with 246/246 ROIs	165/165
ROI series shape (135, 246)	165/165
NaN / Inf / zero-variance	0 / 0 / 0
ReHo outside [0, 1]	0
FC symmetry error	2.2×10^{-16}
DC recomputed from FC	0.000
Consistent scans	165/165

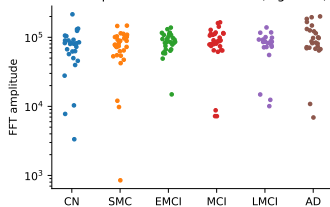
Pilot reproduction gate

Before scale-up, the dataset code had to reproduce the validated pilot: ALFF, ReHo, ROI series and atlas labels came back **bit-identical**.

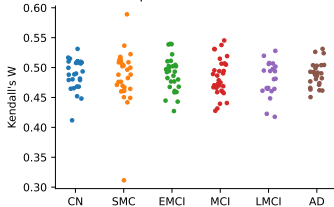
Vectorised ReHo: $\sim 68\text{ s} \rightarrow \sim 3\text{ s}$ per scan.

Results: Why Normalisation and Redundancy Removal

Per-acquisition mean raw ALFF (log scale)

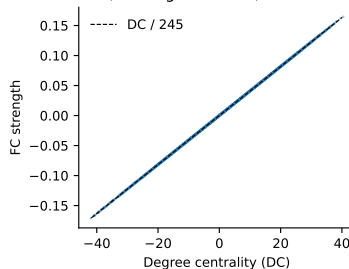


Per-acquisition mean raw ReHo



Raw mean ALFF varies $\sim 250\times$ between scans

40,590 region values, 165 scans

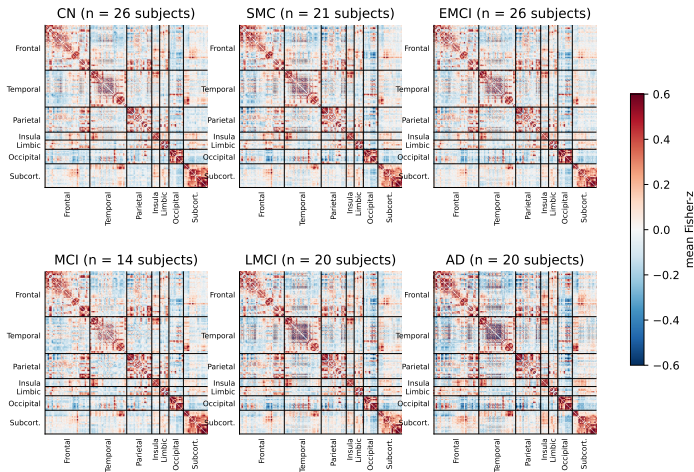


FC strength = DC/245 $\Rightarrow$ removed

After normalisation

mean(mALFF) = mean(mReHo) = std(DC_z) = 1.000000 in 165/165; Fisher-z diagonal exactly 0;
no clipping needed anywhere.

Results: Group-Mean Functional Connectivity



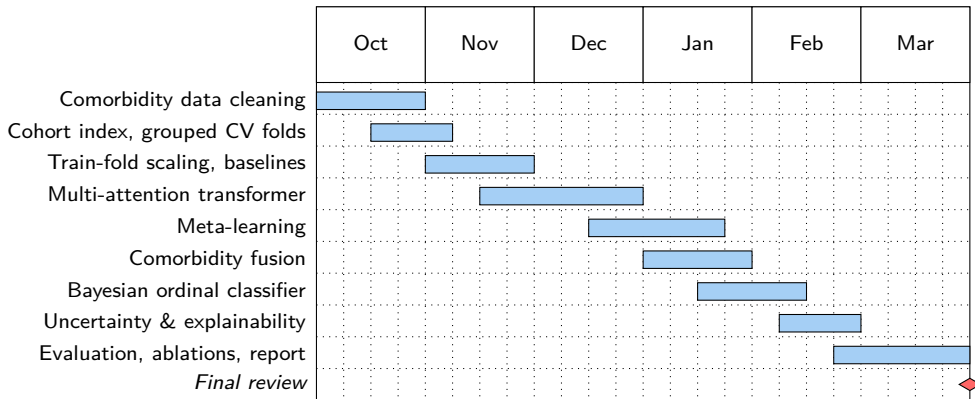
Observation

- ▶ Same structure in all six groups: strong within-lobe blocks, distinct subcortical block
- ▶ Averaged per subject first (no session inflation)
- ▶ **Descriptive only** — discrimination is the model's job

Comparison with a Conventional Pipeline

Aspect	Typical (T1w available)	This project (BOLD-only)
Normalisation	BOLD → T1w → template	direct EPI → MNI (ANTs SyN)
WM/CSF signals	subject tissue segmentation	template priors + reliability gate
Slice timing	corrected from header	not recoverable → not performed
Band vs. TR	one fixed band	Nyquist checked per scan
Verification	toolbox outputs used directly	exact recomputation, 165/165
Leakage control	whole-dataset scaling common	within-subject; subject-grouped CV

Plan for the Remaining Work (tentative)



Team Responsibilities

Team Member	Responsibilities	Current Status
Harishkanna R	[Responsibilities]	[Status]
[Student 2]	[Responsibilities]	[Status]
[Student 3]	[Responsibilities]	[Status]

- ▶ The data foundation is **complete and verified**: 165 scans, 127 subjects, six stages.
- ▶ Every BOLD-only limitation is handled by documented exclusion or substitution — **nothing fabricated**.
- ▶ Review-I design corrected: 246×3 features, **subject-level** validation.
- ▶ Next: comorbidity-aware Meta-Bayesian ordinal model with subject-grouped evaluation.

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Thank You

Questions and Discussion